Audio Transcription Sample

Sourse: Security Catalyst(How the Internet works)
Transcribed by: Karen Christina
Style: Clean Verbatim with Timestamps

[00:00 - 01:00] Introduction and Internet “Myth”

So, how does the internet work? Most of us know how to use the internet without actually understanding how it works-sort of like electricity in your home. You use it every day, but may not understand the mechanics behind it. If the electric grid is difficult to understand, then the internet must be impossible, right? Wrong! In the next few minutes, I’ll put you in the top 10 percent of people understand how the internet actually works. For Security Catalyst.com ,I am Aaron Titus.

Whenever most people think of the internet, this is what comes to mind. The internet is not a bubble cloud, even in the new age of cloud computing the whole fuzzy cloud picture was created by people more concerned about job security than education. This is the internet, the internet is a wire, actually buried in the ground. It might be fiber optics,copper or occasionaly beamed to satellites or through cell phone networks, but the internet is simply a wire.

[01:01 - 02:15]Servers, Clients, and IP Address

The internet is useful because two computers connected directly to this wire can communicate. A server is a special computer connected directly to the internet and web pages or files on that server’s hard drive. Every server has a unique Internet Protocol address or IP address just like a postal address. IP addresses help computer find each other. But since 72.14.205.100 it doesn’t exactly roll off the tongue, we also give them names like google.com, facebook.com or security catalyst.com. So, this is how it works. Your computer at home is not a server because it's not connected directly to the internet. Computers you and I use every day are called clients because they are connected indirectly to the internet through an Internet Service Provider. Here we'll pretend that this is my home laptop and I am using DSL. Now, let's pretend that I want to visit aol.com which is coincidentally both a server and an ISP. I hope onto my laptop with DSL go through my ISP onto the internet, and look at aol.com. My computer connects with aol.com and I can look at its web pages.

[02:16 - 03:30] How Emails and Data Packets Travel

Now, let's say that I want to send an email to Aunt ruth. Aunt ruth has AOL dial-up from home, and I've got a Gmail account. I log onto gmail.com and compose a message to Aunt ruth's email address, auntruth@aol.com. Once I click send, gmail.com sends the email to aol.com. The next day, Aunt ruth dials into AOL servers and retrivers the email.

Whenever an email, picture or web page travels across the internet, computers break the information into smaller pieces called packets. When information reaches its destination , the packets are ressembled in their original order to make a picture, email, web page or tweet.

[03:31 - 04:30] Routers and Packets Wrapping

Okay, So imagine you are at work sitting next to your boss, and you are both surfing online. Your boss is doing market research, and you are updating your facebook profile. You both sending packets back and forth over the internet, but what is to keep your packets from accidently ending up on your boss's screen?
That could be embarrassing!

The solution to the problem is IP addresses and routers. Everything connected directly or indirectly to the internet has an IP address everything! That includes your computer, servers, cell phones and all of the equipment in between. Anywhere two or more parts of the internet intersect, there's a piece of equipment called a router. Routers direct your packets around the internet, helping each packet get one step closer to its destination. Every time you visit a website, upwards of 10 to 15 routers may help your packets find their way to and from your computer.

Imagine each packet as a piece of candy wrapped in several layers. The first layer is your computer's IP address. Your computer sends the packet to the first router, which adds its own IP address. Each time the packet reaches a new router, another layer is added until it reaches the server. Then, when the server sends back information, it creates packets with an identical wrapping. As the packet makes its way over the internet back to your computer, each router unwraps a layer to discover where to send the packet next until it reaches your computer, and not your boss's.

[04:31 - End] Conclusion

And that's how the internet works in five minutes or less and you are now easily in the top 10 percent of people who understand the basics of the internet. If you found this video helpful, check out security catalyst.com for all kinds of ideas on how to protect your information. I am Aaron titus.